CS5100 – Foundations of Artificial Intelligence | Spring 2026 | Prof. Jonathan Mwaura

K-Means Clustering: An Experimental Reproduction Study

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FAI Capstone – Final Report (Milestone 4)

Demo Video: Video Link
GitHub Repository: Github Link

Abstract

The following report documents the implementation of an experimental simulation of the K-Means clustering algorithm that was initially proposed by MacQueen (1967). The algorithm was built completely from scratch using Python programming language and augmented by the use of the following techniques: Elbow Method and Silhouette Analysis (Rousseeuw, 1987) for choosing the optimal number of clusters, K-Means++ initialization scheme (Arthur & Vassilvitskii, 2007), PCA for visualization, and K-Medians clustering as an extension added based on the results of Milestone 3. Results of experiments performed on UCI Iris dataset show that there are three optimal clusters with K-Means++ providing better stability than random initialization. The paper is classified as fully reproducible.

1. Introduction

Clustering is one of the basic tasks performed in unsupervised learning by machines. In this case, data samples are grouped in such a way that similar data samples belong to the same cluster without having to label them in advance. One of the most popular clustering algorithms is the K-Means clustering technique, which was proposed by MacQueen in 1967.

However, there are at least three major drawbacks of the K-Means method. First, the number of clusters has to be known in advance. Second, initial placement of centroids can influence the outcome significantly. Third, data samples in multiple dimensions cannot be viewed visually. This project reproduces the core algorithm from scratch and extends it with targeted solutions to each problem.

Keshvi focused on optimal k selection using the Elbow Method and Silhouette Analysis. Parthiv focused on initialisation stability using K-Means++ and visualisation using PCA. Both members jointly implemented the baseline K-Means algorithm. Following feedback from the professor on the Milestone 3 submission, we additionally implemented K-Median clustering as a comparison method. We selected this paper because K-Means sits at the foundation of all clustering research. Building it from scratch provides genuine insight into every design decision the algorithm requires.

2. Paper Overview

MacQueen's 1967 paper, Some Methods for Classification and Analysis of Multivariate Observations, formalised the k-means concept and introduced the term itself. The algorithm partitions N-dimensional data into k groups by minimising total within-cluster variance.

Core Idea

The algorithm alternates between two steps: assigning every data point to the nearest centroid by Euclidean distance, and recomputing each centroid as the mean of all assigned points. MacQueen described an incremental version; our implementation uses the standard batch version which is the actual form used in practice.

Main Contributions

The paper formally defined the K-Means objective, proved convergence in finite steps, and demonstrated application to real multivariate data.

Gaps Addressed by This Work

MacQueen's paper does not explain how to choose k, does not handle poor random initialisation, and contains no visual analysis. These three gaps are addressed by our extensions.

3. Reproduction Objective

Our reproduction targeted: (1) Core K-Means algorithm - random initialisation, batch assignment, centroid update, convergence check. (2) Inertia - within-cluster sum of squared distances. (3) Elbow Method - inertia vs k. (4) Silhouette Analysis - from scratch using Rousseeuw's formula. (5) K-Means++ - distance-weighted centroid seeding. (6) PCA visualisation - 4D to 2D. (7) K-Median clustering - added following professor feedback on Milestone 3.

We intentionally excluded the incremental variant from MacQueen's paper and did not use sklearn's built-in KMeans, so every step is transparent and verifiable.

4. Methodology & Implementation

4.1 Core K-Means Algorithm

initialize_centroids: Randomly selects k rows from the dataset without replacement as starting cluster centres.

assign_clusters: Uses NumPy broadcasting - X reshaped to (n,1,d) minus centroids (k,d) gives (n,k,d) differences. np.linalg.norm on axis=2 gives the (n,k) distance matrix; argmin on axis=1 gives the nearest centroid per point.

update_centroids: For each cluster i, filters assigned points and computes column-wise mean. Returns new (k,d) centroid array.

kmeans_scratch: Main loop uses np.allclose for convergence and max_iters= 100 as safety cap.

4.2 Inertia

Within-cluster sum of squared distances - the K-Means objective - computed manually for full transparency.

4.3 Elbow Method

K-Means run for k=1 to 10. The point where the inertia curve flattens is the elbow and indicates the best k.

4.4 Silhouette Analysis

For each point i: a(i) = average distance to same-cluster points; b(i) = average distance to nearest other-cluster points; s(i) = (b(i)-a(i))/max(a(i),b(i)). Computed entirely from scratch with no sklearn functions.

4.5 K-Means++ Initialisation

First centroid chosen randomly; each subsequent centroid chosen with probability proportional to squared distance from nearest existing centroid. Spreads starting centres apart.

4.6 PCA Visualisation

PCA is used to reduce the 4-dimensional Iris data to 2 dimensions so the clusters can be plotted. It is fitted on the full dataset using fit_transform(X) and the same transformation is applied to the centroids using transform(centroids) - this ensures both the data points and centroids are in the same coordinate space.

4.7 K-Median Clustering (Added Following Professor Feedback)

As mentioned by the professor in the Milestone 3 submission feedback, we extended our implementation to include K-Median clustering as an additional comparison method. This was added to the notebook after the Milestone 3 review to demonstrate a deeper understanding of the limitations of K-Means and how alternative centre-update strategies affect clustering behaviour.

K-Median is structurally identical to K-Means, with one key difference: instead of updating each cluster centre using the mean of assigned points, it uses the median. This single change has an important practical consequence - the median is resistant to outliers, whereas the mean is not.

How K-Median Works

(1) initialize_medians: Selects k random rows as starting medians - identical to K-Means initialisation.

(2) assign_clusters_median: Computes Euclidean distance from each point to each median and assigns the point to the nearest one - same logic as K-Means assignment.

(3) update_medians: For each cluster i, computes np.median(cluster_points, axis=0) - the element-wise median across all assigned points. An empty cluster check is included: if a cluster has no points assigned, a random point from the dataset is chosen as the new median to prevent crashes.

(4) kmedian_scratch: Main loop identical to K-Means, using np.allclose for convergence.

K-Means vs K-Median - Key Difference

Property	K-Means	K-Median
Centre update	Mean of cluster points	Median of cluster points
Outlier sensitivity	High - mean is pulled by extreme values	Low - median ignores extremes
Computational cost	Lower	Slightly higher
Best used when	Data is clean, no outliers	Data may contain outliers

In case of Iris dataset, K-Median performs almost similarly as K-Means since the dataset is free from any kind of outliers. The visualization of clusters in PCA using K-Median reveals the formation of three clusters, which have medians lying almost at the same point as K-Means centroids lie. It clearly explains the fact that in the absence of outliers in the data set, both techniques arrive at the same solution, but K-Median should be preferred in case of noisy data sets.

5. Experiments

5.1 Dataset

Property	Details
Dataset	UCI Iris
Samples	150
Features	4 (sepal length, sepal width, petal length, petal width)
Known Classes	3 (Setosa, Versicolor, Virginica - 50 each)
Preprocessing	StandardScaler before PCA; raw features used for K-Means and K-Median

5.2 Experiment 1 - Elbow Method

K-Means run for k=1 to 10, inertia recorded and plotted.

5.3 Experiment 2 - Silhouette Analysis

k=2 to 10 tested; average silhouette score computed from scratch for each k.

5.4 Experiment 3 - Initialisation Stability

Random initialisation and K-Means++ each run 10 times for k=3. Mean and Standard deviation of silhouette score compared.

5.5 Experiment 4 - PCA Visualisation

K-Means++ final clustering (k=3) projected to 2D via PCA and plotted.

5.6 Experiment 5 - K-Median Comparison

K-Median run with k=3 on the same Iris dataset. Cluster assignments and median positions visualised via the same PCA projection used for K-Means, enabling direct visual comparison.

6. Results

6.1 Elbow Method

Table 1 shows inertia for k=1 to 10. The elbow is clearly visible at k=3.

Table 1. Inertia vs Number of Clusters

K	Inertia	Observation
1	681.37	All points in one cluster - very high spread
2	152.35	Large drop - two groups separated
3	78.85	Clear elbow - drop slows significantly → Optimal
4	71.77	Diminishing returns begin
5	46.47	Continues to decrease slowly
6-10	42-30	Very small marginal gains

6.2 Silhouette Scores

Table 2 shows silhouette scores. Silhouette peaks at k=2 but k=3 is the correct choice supported by Elbow and domain knowledge.

Table 2. Silhouette Score vs Number of Clusters

K	Score	Observation
2	0.6810	Highest score - Setosa cleanly separated from the rest
3	0.5185	Strong score - three distinct clusters → Final choice
4	0.4963	Noticeable drop
5-10	0.38-0.28	Gradual decline

6.3 Initialisation Stability

With random initialisation, the starting centroids sometimes fall too close together within the same natural cluster. It makes the algorithm misclassify two actual clusters into one cluster. The resulting classification is different for every seed number.

K-Means++ solves this problem by picking up centroids that are as distant from already picked centroids as possible. This spreads the starting points across different regions of the data, giving the algorithm a much better starting point.

In our experiments on the Iris dataset, K-Means++ consistently found the correct three-cluster structure matching the known species, while random initialisation occasionally converged to a wrong grouping. This confirms the main claim of Arthur & Vassilvitskii (2007) - careful seeding leads to more reliable results.

6.4 PCA Visualisation

The scatter plot shows three reasonably separated clusters. Setosa is tightly grouped and cleanly isolated. Versicolor and Virginica overlap slightly, consistent with their biological similarity. Centroids sit at the geometric centre of each cluster, confirming correct convergence.

6.5 K-Median Results

K-Median with k=3 produced cluster assignments very similar to K-Means on the Iris dataset. The PCA plot of K-Median clusters shows the same three-group structure. The median positions are close to but not exactly identical to the K-Means centroids - the median is computed element-wise and pulls slightly less toward any extreme points. The general clustering result from both algorithms appears to be quite similar due to the absence of outliers.

Table 4. K-Means vs K-Median Comparison (k=3, Iris Dataset)

Property	K-Means	K-Median
Cluster structure	3 groups matching species	3 groups - same structure
Centre type	Mean of cluster points	Median of cluster points
Outlier robustness	Lower	Higher
Convergence speed	Faster	Slightly slower

7. Analysis and Discussion

7.1 Support for MacQueen's Claims

Our experiments confirm MacQueen's claims: the algorithm consistently converges in well under 100 iterations, inertia decreases monotonically within each run, and the three discovered clusters align closely with the known Iris species.

7.2 Why Silhouette Peaks at k=2

Setosa is very clearly separated from the other two species. From a pure distance perspective two clusters score highest. However, the Elbow Method and domain knowledge both confirm k=3. Combining methods gives the correct final answer.

7.3 Impact of Initialisation

K-Means++ consistently produced the correct three clusters on the Iris dataset, while random initialisation occasionally converged to a wrong grouping due to poorly placed starting points - confirming that careful seeding leads to more reliable results.

7.4 PCA Interpretation

The PCA plot shows three reasonably separated clusters with centroids clearly visible, and some overlap between Versicolor and Virginica reflecting the natural complexity of the dataset.

7.5 K-Median vs K-Means

On the clean Iris dataset, K-Median and K-Means produce nearly identical cluster structures. This is expected - median and mean converge to similar values when data is symmetric and outlier-free. The real value of K-Median would be apparent on noisier datasets where outliers pull K-Means centroids away from the true cluster centre. This extension demonstrates that understanding the algorithm deeply enough to swap the update rule is itself a meaningful exercise in comprehending what K-Means actually does.

8. Challenges in Reproduction

(1) Incremental vs batch: MacQueen describes an incremental rule. We used batch and documented the justification.

(2) Convergence criterion: No stopping condition in the original paper. np.allclose with standard tolerances was chosen.

(3) Choosing k: Neither Elbow nor Silhouette alone gave an unambiguous answer. Both methods plus domain knowledge were needed.

(4) Silhouette scalability: Scratch implementation is $O(n^2)$. Fine for 150 samples; problematic at larger scale.

(5) PCA consistency: Applying fit_transform to centroids separately created an inconsistent coordinate space. Identified and fixed.

(6) K-Median empty cluster: When k is large relative to data size, some clusters can become empty during iteration. We handled this by reassigning a random data point as the median when an empty cluster is detected.

9. Limitations

This study used only the UCI Iris dataset (150 samples). Behaviour on larger or noisier data may differ. The scratch Silhouette implementation is not optimised for scale. K-Means assumes spherical equally-sized clusters. Only Euclidean distance was used. Ten random seeds provide reasonable but not exhaustive stability estimates. K-Median's advantage over K-Means could not be fully demonstrated on the clean Iris dataset - a dataset with outliers would provide a more compelling comparison.

10. Conclusion and Future Work

We successfully reproduced the K-Means algorithm from scratch and extended it with Elbow Method, Silhouette Analysis, K-Means++ initialisation, PCA visualisation, and K-Median clustering. Our experiments confirm k=3 as optimal, K-Means++ as significantly more stable than random initialisation, and both K-Means and K-Median as recovering the correct Iris species groupings.

The K-Median extension, added following professor feedback on Milestone 3, reinforces a key insight: K-Means is simple in concept but understanding it deeply means knowing exactly which design choices can be changed and what effect those changes have.

Future work: Test on datasets with real outliers to better demonstrate K-Median's advantage; test on larger datasets; implement the incremental variant and add alternative distance metrics.

11. Reproducibility Verdict

Verdict: Fully Reproducible

MacQueen's K-Means algorithm is fully reproducible from the original 1967 paper. All three extensions are likewise reproducible from their respective source papers. The K-Median extension is a straightforward algorithmic variant with no reproducibility concerns.

12. Member Contributions

Member	Contributions
Keshvi Choksi	Baseline K-Means (joint), Elbow Method, Silhouette Analysis from scratch, K-Median implementation (joint), report sections 1, 2, 3, 6.1, 6.2, 8, 9
Parthiv Modi	Baseline K-Means (joint), K-Means++, PCA pipeline, stability experiments, K-Median implementation (joint), report sections 4, 5, 6.3–6.5, 7, 10, 11

References

[1] J. MacQueen, "Some Methods for Classification and Analysis of Multivariate Observations," in 1967.

[2] D. Arthur and S. Vassilvitskii, "K-Means++: The Advantages of Careful Seeding," in 2007.

[3] P. J. Rousseeuw, "Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis," in 1987.